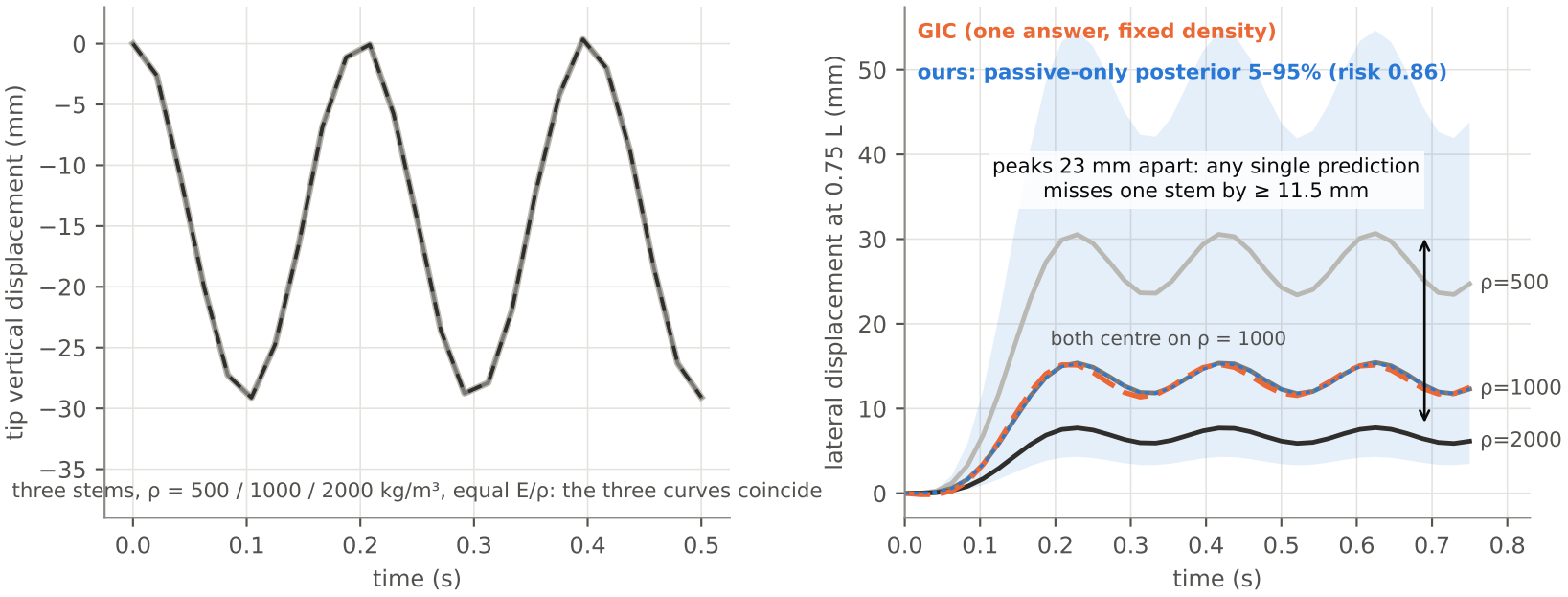
What the cameras see: passive sag
tip vertical displacement (mm)
0
−5
−10
−15
−20
−25
−30
−35
0.0
0.1
0.2
0.3
0.4
0.5
time (s)
three stems, ρ = 500 / 1000 / 2000 kg/m³, equal E/ρ: the three curves coincide
What a robot needs: response to a 20 N push
GIC (one answer, fixed density)
ours: passive-only posterior 5–95% (risk 0.86)
peaks 23 mm apart: any single prediction
misses one stem by ≥ 11.5 mm
both centre on ρ = 1000
ρ=500
ρ=1000
ρ=2000
lateral displacement at 0.75 L (mm)
50
40
30
20
10
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
time (s)